PLANT ROOM DRAINAGE & SUBMERSIBLE DEWATERING DETAIL:

1. Perimeter floor drainage trenches in Chiller room, Boiler room, Substation, and DG yard with heavy-duty ductile iron gratings.
2. Sump pit construction: 3500 x 3000 x 2800 mm deep reinforced concrete pit with crystalline waterproofing compound.
3. Dewatering pump set: Dual submersible non-clog cutter pumps (1 Duty + 1 Standby) each rated for 50 m3/hr @ 18m head.
4. Pump impeller designed to pass solid spherical debris up to 50mm diameter without clogging.
5. Automated 4-float level switch arrangement: Low Cut-off, Lead Pump Start, Lag Pump Start (High Inflow), and Critical High Alarm.

Sump Pit Tag	Location / Area	Pit Dimensions	Discharge Pump	Flow Rate	Discharge Lin
SUMP-01	Chiller Plant Room	3.5 x 3.0 x 2.8m	Dual Submersible 5.5 kW	50 m3/hr @ 18m	100mm CI Pipe
SUMP-02	Main Substation Trench	2.5 x 2.0 x 2.2m	Dual Submersible 3.7 kW	30 m3/hr @ 15m	80mm CI Pipe
SUMP-03	DG Fuel Tank Yard	2.0 x 2.0 x 2.0m	Flameproof Submersible 2.2 kW	15 m3/hr @ 12m	65mm GI Pipe
SUMP-04	Fire Pump Room	3.0 x 2.5 x 2.5m	Dual Submersible 5.5 kW	50 m3/hr @ 18m	100mm CI Pipe

AMARAVATI GOVERNMENT COMPLEX · LEGISLATIVE ASSEMBLY		
DWG TITLE: BASEMENT MEP PLANT ROOM FLOOR DRAINAGE & SUMP		
DRAWING NO: SL-288-DRF-101	REV: R1	SCALE: 1:100
CLIENT: AP CRDA / GOVT OF ANDHRA PRADESH	DATED: 28-09-2026	